AI-Based Smart Civic Issue Detection and Complaint System

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ABSTRACT: We are seeing an increase in civic Issues like potholes, waste management, waterlogging, and insufficient street facilities. Old complaint systems often require a lot of manual effort, leading to some uncertainty in data, slow response times, and Lack of priority planning. This paper introduces CivicAI, an AI-powered complaint reporting system. It generates It writes the complaint automatically, gets the location, and gives a unique complaint number for each issue. The system Backend works using Node.js and Express.js, and users see a simple webpage built with HTML, CSS, and JavaScript. Admins can use a dashboard to check complaints and see whether they are solved or not. The system works correctly, collects information well, and responds quickly, so it is useful for real-life civic issues. The system safely connects people and authorities and helps solve civic problems faster using automation.

KEYWORDS: AI, Gemini AI Model, Complaint Handling System, Image Recognition, Language Understanding

1.INTRODUCTION

Due to Urbanization the problems of civic issues that local authorities face, including potholes, waste accumulation, waterlogging, and broken streetlights[1,2]. To Protect our City infrastructure and Keep Citizen satisfied , Civic problems Should be Reported And Solved Efficiently. Most Existing Systems Are Manual, Slow And Hard to Use this systems. In Previous System Citizens have Fill Long Form or Make long Calls, Which leads to Wrong Information. Sometimes There is No Fixed Categories and Perfect Information Of that Problem[4]. When Like This Problem Comes It Leads To Delay For Solving the Problem The Faster growth of AI and Computer Vision offers better Ways To Solve a problem there are Many Ways To Solve a Problem by Using AI and Computer Vision . By Using AI The System Gives Us the Problem in image.

CivicAi uses AI to analyze and Solve The problems. Users Upload The Photo Of Problem And the System Processes It Using Gemini AI to Identify the problem Such as Garbage, Potholes , Etc. By Using Gemini

AI in System it Automatically writes the complaint, take the Live Location and Assigns A Unique Id To Track The Status Of Complaint .The System Backend Works On Node.js and Express.js,while the Frontend Built By Using HTML,CSS and Also JavaScript .To Manage The Complaints Easily The Admin Uses Dashboard. This paper Tells Us About How Civic Ai Uses To Improve the Connections and AI Also Helps The System to Manage Civic Problem Faster And in Accurate Way.

2.ADVANTAGE OF PROPOSED MODEL

The proposed CivicAI model is useful because it makes the complaint process easier for normal people. In traditional systems people have to write long descriptions and many times the information is not clear. In this user can just upload a photo and submit the issue which saves time and effort.

Another advantage about this system is that it automatically connects the complaint with the location. This helps authorities understand where the problem is without confusion. Since complaints are saved properly the same issue does not get repeated again and tracking also becomes simple.

CivicAI is simple to use and does not require filling long forms. Users only need to upload a photo to report a problem which makes the system easy even for people with less technical knowledge. The admin side is controlled and secure allowing only authorized people to manage and update complaints properly..

3.SYSTEM ARCHITECTURE

CivicAI is an AI-based civic help system that is mainly developed to make the process of reporting public problems easier for people and simpler to manage for authorities. The system is designed in a layered manner so that every part has its own responsibility and the overall system works smoothly without becoming complicated.

The Presentation Layer is where users actually interact with the system. Through a simple web interface, citizens can log in, upload images of civic issues, and check what is happening with their complaints. Administrators also use this interface to view complaints, manage records and observe common problems occurring in different areas. This layer mainly focuses on making the system easy to use and understand.

The Application Layer handles the internal working of CivicAI. It takes requests from users, verifies them, and processes the complaints properly. It also makes sure that basic security rules are followed. The data layer is used to store all system related information. User details, complaint records, and uploaded images are saved in an organized manner so that they can be accessed or updated whenever required. Proper data storage helps avoid duplication and keeps the system running without confusion.

The external services layer is used to improve system functionality by connecting with external tools. The Google Gemini API is used to analyze uploaded images and identify the type of issue shown in them. Location services are also used to capture the area where the problem is reported, which helps in making complaints more accurate[3,7].

Because of this layered approach, CivicAI remains easy to maintain, flexible for future changes, and capable of handling more users.

4.RESEARCH METHODOLOGY

Image recognition model:the civicai system uses image recognition model to understand and analyze the uploaded captured image by the user by using techniques such as advanced computer vision.the system uses google gemini's vision ai model for understanding real world images by using deep learning techniques. the gemini vision model takes an image as an input such as ex. image of a pothole on a road ,image of broke/fallen/non-working streetlight, image showing dumped garbage on road,the gemini vision model then analyses the image to classify the image into its specific pre-defined category which best describes the civic issue in the image[13,14].after analyzing the image the model gives a predicted category for civic issue and its confidence score.the images which got lower confidence score are highlighted for admin so that the admin can later manually review the images which got lower confidence score from the gemini vision model.

Geo-location capture model:the civicai system uses a geolocation capture model which helps in either getting the live GPS(global positioning system) co-ordinates of the user while filing a complaint or the user can manually select the location for the complaint through the GUI(graphical user interface) of the maps.the system uses browser-based geolocation api to get user's current latitude and longitude once the use grants or allows location access permission[3].

Auto complaint description model:for creating a complete accurate text description of the civic complaint by analyzing the uploaded image and location details the civicai system uses google gemini's nlp(natural language processing) unit which can generate accurate complaint description of civic issue just by analyzing the image and location details. for ex.user uploads an image of a pothole and allows location acces, then the system can create an accurate text description of the complaint such as "a clear pothole is detected on mahatma Gandhi road which puts citizens life in threat so please take necessary action to resolve the issue as early as possible" the description also includes the complaint category for the civic issue. this auto complaint description model helps in quickly creating a complaint description for the complaint which new users can find helpful as no manual description is required. there by making the complaint filing process more simpler and easier for new users who don't have any much experience of filing a complaint[5,8].

5.SYSTEM WORKFLOW

the system consists of the image uploading unit, location co-ordinates capturing unit and an ai analysis unit which gives a predicted class and a confidence score and creates a text description of the complaint. the system is used by two main users one is the citizen who will file the complaint and another is the admin who will review and forward the complaints to the respective department admin. the workflow of the system starts from a citizen creating and logging into his user account then the citizen upload an image of the civic issue and gives his location access then ai analysis happens which returns a predicted complaint category and its confidence score and a text description of the complaint then citizen can file the complaint .once the complaint is filed by the citizen the complaint gets visible to the admin who then reviews the complaint and then forward the complaint to the respective responsible department admin who is responsible for resolving the complaint[1,3,10,12].

6.RESULTS AND DISCUSSION

This system CivicAI was created and then tested for reporting civic issues and ease of the registration of complaint by the automating it. The test conducted gave accurate results of function, accuracy of

geolocation, how fast the response is returned, if the interface is user friendly or not. Even the admin dashboard is evaluated so that it can function well. We have used a class of images to test the system. It includes images of water logging, broken street lights, Roadkill, garbage stagnation, Potholes. Gemini Vision AI was able to accurately categorize 91 out of the 100. This meant that 91 percent was correct response. The rest varied due to slight issues like lighting was less and some objects were not properly visible. The automatically text generated feature was proved very essential and easy because it automatically generated short and concise description of the issue. This saved time that would be required to type it manually. In our system, the positioning manual very accurately positioned the location of the user within 10 meter radius of the reported issue, hence the accurate location was able to be captured.

Overall, it took 3 to 5 seconds from uploading image to submitting the complaint, which is very fast. When checking the admin dashboard we saw that the review of complaint lodged was visible on the dashboard and it was easily manageable. Feedbacks from the user showed us that our system is simple to use and effective as well. The most appreciated part of thw system is that all the issues were resolved with just one tap.

Overall the system is feasible and robust [9]. The CivicAI system integrates web technology, Artificial Intelligence and Geolocation to successfully record, manage and analyze complaints effectively. This discussion makes it clear that using AI manual workload gets eased and there is transparency between both ends. We are going to improve the system by adding features like database migration, multilingualism, Duplicate image detection, etc.[4,8,13]

7.CONCLUSION AND FUTURE SCOPE

CivicAI system shows how Artificial Intelligence can be used to manipulate and alter already existing softwares in a good way so that the software can be updated and automated. This reduces the need of human engagement and the risk of errors get reduced too. We have integrated Gemini Flash 2.5, The issues get detected easily because the API has computer vision which accurately detects issue just by using image as an input. The images are categorized into the types of civic problems like Potholes, waterlogging, Roadkill, etc. The user interface is very easy and simple since it is internet based. Hence it becomes easy to add and remove admin. The tests showed that the location captured are highly accurate because of the Artificial Intelligence's capacity to capture highly accurate site coordination.

In conclusion, CivicAI is easy to expand and user friendly. This system will help the citizen connect with the administration without the haste of the complications that exist right now due to which there is no proper way to establish communication between the governance and the citizens. This study has contributed towards the development of smart governance. There remains room for improvement such as integration of database, multilingualism, etc. Advancement in these regions can help us bridge gap between the administration and civilians. Future scope could be using more advanced database like MongoDB, adding multiple languages so that the user from different backgrounds can easily accessible the software, analyse duplicate records, Sending updates to the citizen through mails amd phone number. These updates will help make CiviAI more smart.

Table 1. Key Features and Applications of CivicAI System

S No	Feature	Description	References
1	AI-Powered Image Classification	Automatically categorizes civic issues (potholes, waste, waterlogging, streetlights) using Gemini Vision AI with 91% accuracy	[3,7,12]
2	GPS-Based Geolocation	Captures real-time location with <10m error margin for accurate issue mapping and rapid response deployment	[3,10]
3	Automated Complaint Generation	Uses NLP to generate structured complaint descriptions, eliminating manual form filling and ensuring standardization	[5,8]
4	Admin Dashboard	Provides real-time complaint monitoring, status updates, duplicate detection, and task assignment for authorized personnel	[1,4]
5	User-Friendly Interface	Single-tap image upload system built with HTML, CSS, JavaScript for accessibility across all user demographics	[6,11]
6	Scalable Architecture	Node.js/Express.js backend with JSON storage, designed for easy migration to MongoDB/PostgreSQL for enterprise deployment	[6,8,14]

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